

Beat Transport Challenge

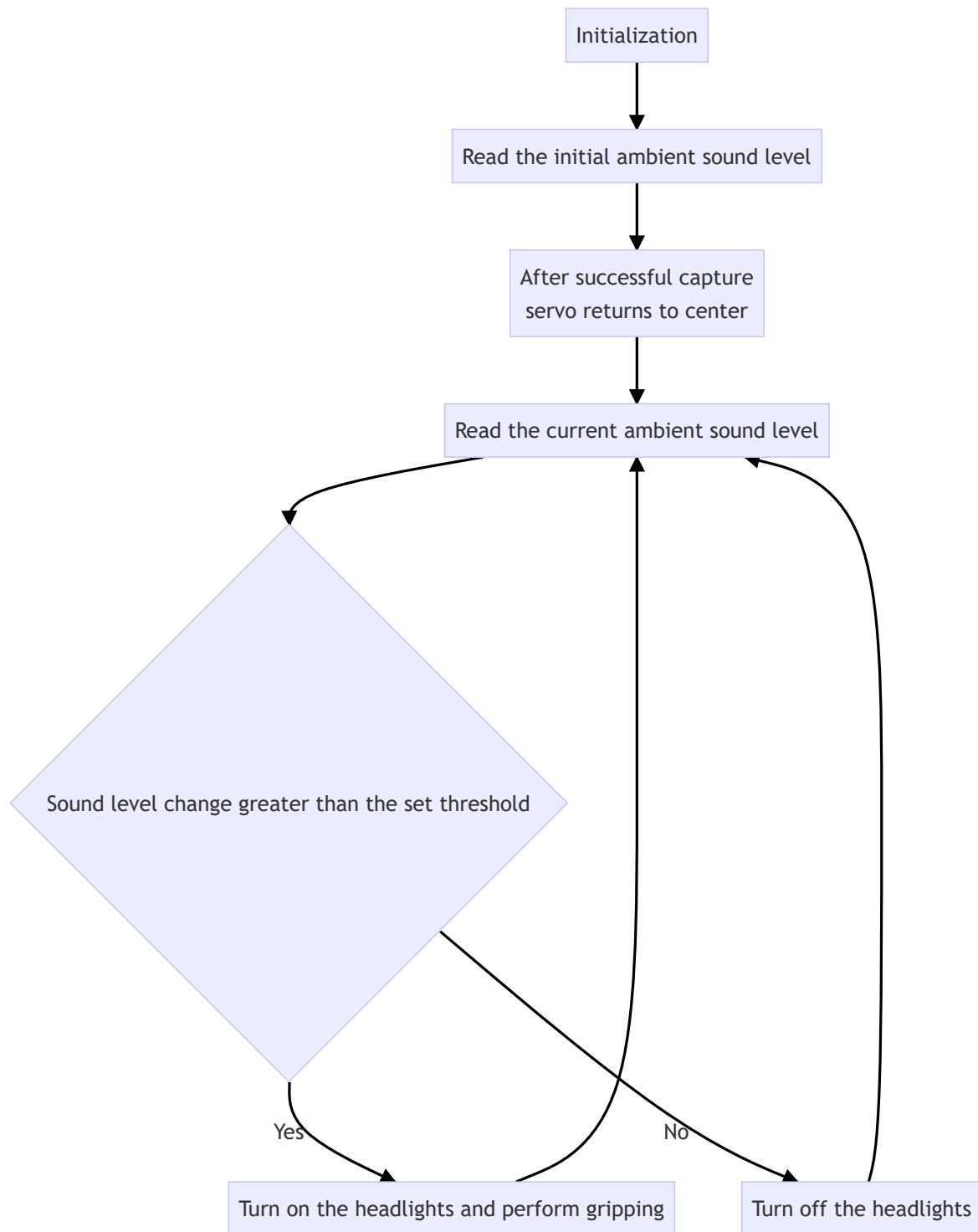
Beat Transport Challenge

1. Experiment Purpose
2. Implementation Principle
3. Experiment Steps
4. Experiment Steps
5. Experiment Phenomenon

1. Experiment Purpose

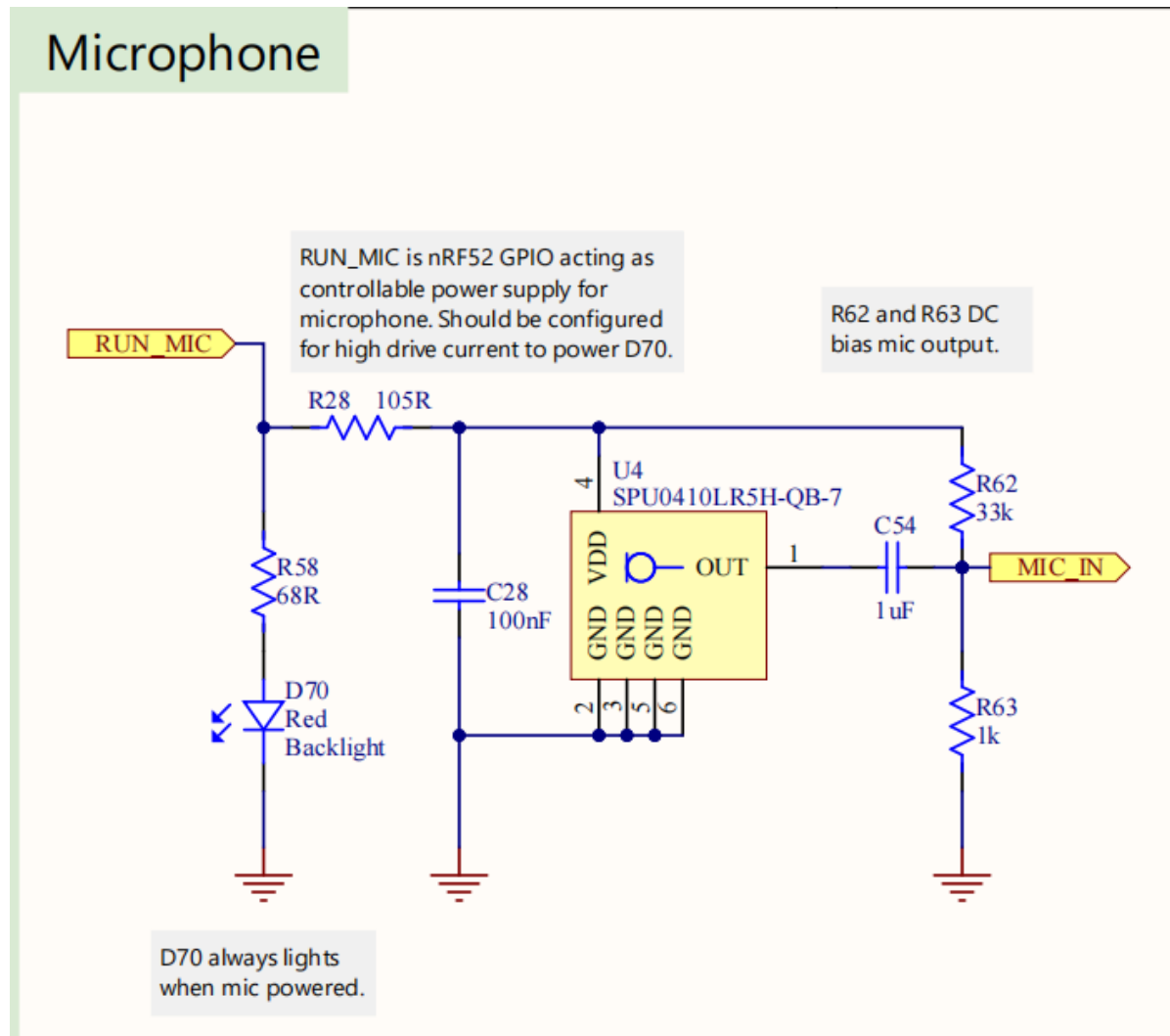
In this section, we will use the sound sensor together with servo control to achieve the "beat transport challenge" effect.

2. Implementation Principle

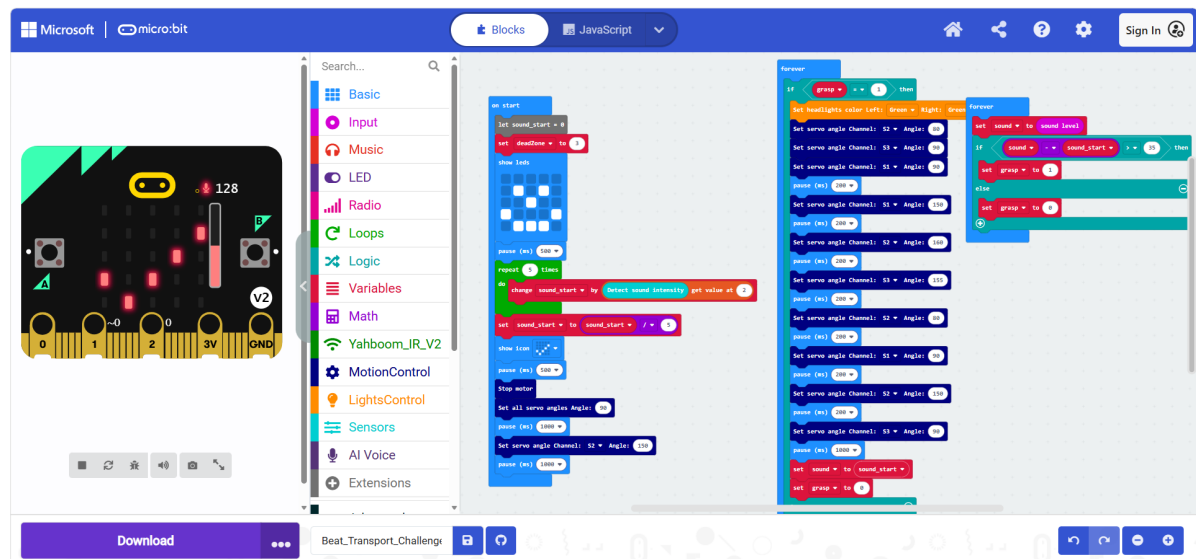


V2 version (Micro:bit V2 built-in microphone)

A microphone sensor is provided near the Logo of the Micro:bit V2. When the program drives and reads it, the red LED lights up; when it is not read, the red LED turns off and the microphone is shut down, which can reduce the power consumption of the Micro:bit at the hardware level.

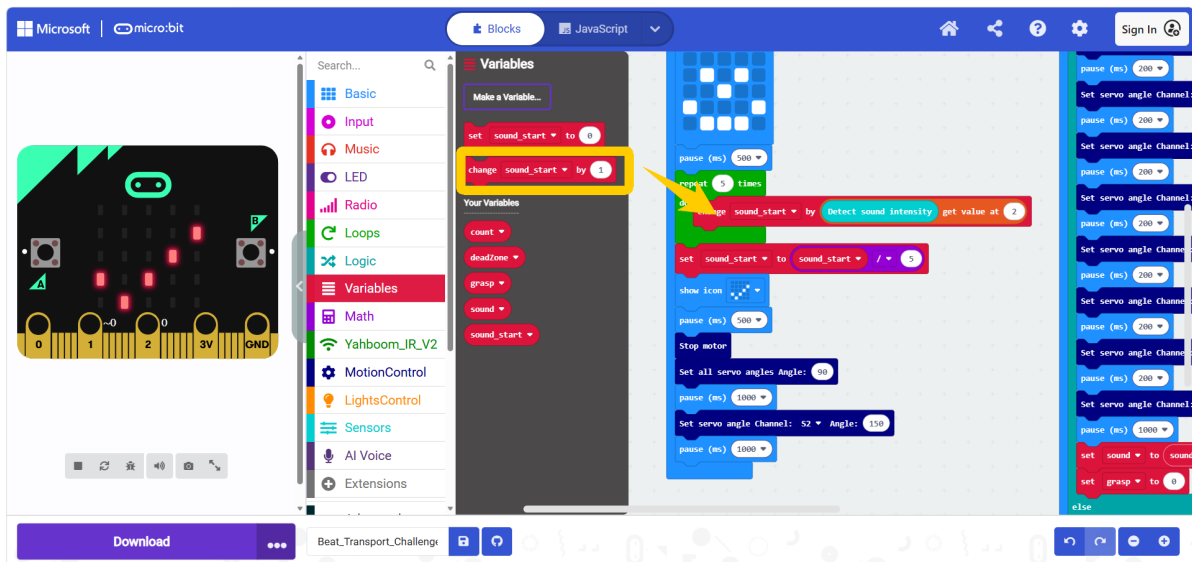


V2 version program



V1V2 version (expansion board MIC sensor)

The on-board sound sensor consists of an electret microphone (commonly called a mic head) plus signal amplification and clamping protection, forming a complete sound detection circuit.



The positions of the blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

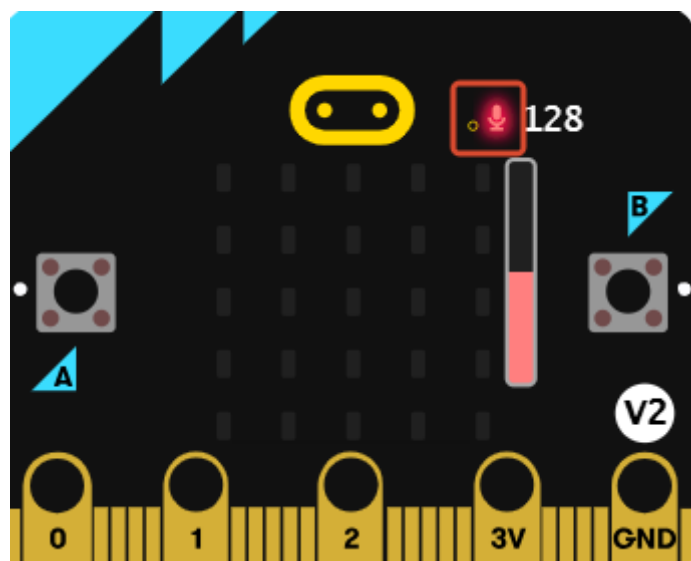
1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `15.Beat_Transport_Challenge_v1v2.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

After flashing and turning on the power, first place the cube EVA block in the middle of the gripper. When a change in the ambient sound is detected, the spotlight turns on green and the car begins to grip the block. **The preset sound threshold can be adjusted according to the actual situation.**

After the transport is complete, turn off the lights.

The recording position of the V2 version is as follows.



The recording position of the V1V2 version is as follows.

